PROJECT REPORT

TRAFFIC LIGHT

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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Sahil Chauhan

ABSTRACT

Traffic congestion has become a major issue in urban areas. Efficient traffic management is essential for reducing delays and improving road safety. This project presents a Traffic Light Controller designed using Verilog HDL and implemented using Xilinx ISE/Vivado.

The controller follows a Finite State Machine (FSM) approach to switch traffic signals automatically in a predefined sequence. The design consists of Red, Yellow, and Green signals for two roads operating alternately.

The project demonstrates digital design concepts such as sequential logic, state machines, clock-driven operation, and FPGA implementation. Simulation is carried out using Xilinx tools to verify the functionality of the controller before hardware implementation.

1. INTRODUCTION

Traffic signal controllers are essential components of modern transportation systems. They regulate the movement of vehicles at road intersections by controlling Red, Yellow, and Green lights.

Traditionally, traffic lights were controlled using relay circuits or microcontrollers. With the advancement of digital electronics, Field Programmable Gate Arrays (FPGAs) provide a faster, more reliable, and flexible solution.

In this project, a Traffic Light Controller is designed using Verilog HDL, synthesized in Xilinx ISE/Vivado, and verified through simulation.

The controller automatically changes signal states according to a predefined timing sequence using a Finite State Machine.

2. OBJECTIVES

The main objectives of the project are:

Design an automatic Traffic Light Controller using Verilog HDL.

Implement the controller using Xilinx ISE/Vivado.

Develop the system using Finite State Machine (FSM).

Simulate the design and verify correct operation.

Reduce human intervention in traffic signal control.

Demonstrate FPGA-based digital system design.

Traffic Light Controller Using Verilog HDL

3. BLOCK DIAGRAM

```
          +----------------------+
          |     Clock Input      |
```

```
          +----------+-----------+
                   |
                   v
          +---------------------+
          |     Reset Button    |
          +----------+-----------+
                   |
                   v
          +---------------------+
          | Verilog FSM Controller|
          +----------+-----------+
                   |
     +------------------+------------------+
     |                                     |
     v                                     v
+------------------+              +------------------+
| Road A LEDs      |              | Road B LEDs      |
| Red Yellow Green |              | Red Yellow Green |
+------------------+              +------------------+
```

Block Description

The Traffic Light Controller consists of four main blocks:

Clock Input: Provides timing pulses for synchronization.

Reset: Initializes the controller to its default state.

FSM Controller: Controls the sequence of traffic lights using Verilog HDL.

LED Outputs: Represent the Red, Yellow, and Green lights for both roads.

4. HARDWARE & SOFTWARE REQUIREMENTS

Hardware

FPGA Development Board

LEDs (Red, Yellow, Green)

Push Button (Reset)

Connecting Wires

Power Supply

Computer/Laptop

Software

Xilinx ISE Design Suite / Vivado

Verilog HDL

ISim or Vivado Simulator

Windows Operating System

5. WORKING PRINCIPLE

The controller operates using a Finite State Machine (FSM).

Initially, Road A receives the Green signal while Road B remains Red.

After a predefined number of clock cycles, the controller changes:

Road A → Yellow

Road B → Red

Then,

Road A → Red

Road B → Green

Again, after the timer expires,

Road B → Yellow

Finally,

Road B → Red

Road A → Green

This sequence repeats continuously.

The design is synchronized using the system clock and can be restarted at any time using the reset signal.

6. FINITE STATE MACHINE (FSM)

```
+-----------+
|  S0       |
| A=Green   |
| B=Red     |
+-----+-----+
      |
      v
+-----------+
|  S1       |
| A=Yellow  |
| B=Red     |
+-----+-----+
      |
      v
+-----------+
|  S2       |
| A=Red     |
| B=Green   |
+-----+-----+
      |
      v
+-----------+
|  S3       |
| A=Red     |
```

```
        | B=Yellow  |
        +-----+-----+
              |
              |
              +-------------+
                            |
                            v
                        S0 Again
```

State Description

State

Road A

Road B

S0

Green

Red

S1

Yellow

Red

S2

Red

Green

S3

Red

Yellow

7. ALGORITHM

Start the system.

Initialize all signals.

Reset the FSM to State S0.

Turn ON Green light for Road A.

Wait for the timer.

Change Road A to Yellow.

Wait for the timer.

Turn ON Green light for Road B.

Wait for the timer.

Change Road B to Yellow.

Return to State S0.

Repeat the cycle continuously.

8. CIRCUIT DESCRIPTION

The controller is implemented on an FPGA using Verilog HDL.

The FPGA receives a clock signal.

A reset switch initializes the controller.

Six output pins drive six LEDs:

Road A: Red, Yellow, Green

Road B: Red, Yellow, Green

Each state transition is triggered by the system clock.

Timing can be adjusted by modifying the counter value

11. XILINX IMPLEMENTATION

Steps

Open Xilinx ISE Design Suite or Vivado.

Create a new Verilog project.

Add the traffic_light_controller.v file.

Add the traffic_light_tb.v testbench.

Run Syntax Check.

Run Behavioral Simulation.

Observe waveform outputs.

Perform Synthesis.

Generate the RTL schematic.

Generate the bitstream and download it to the FPGA (if hardware is available).

12. RTL SCHEMATIC

The RTL schematic generated by Xilinx consists of:

Clock Input

Reset Logic

Finite State Machine

State Registers

Combinational Logic

LED Output Drivers

The schematic verifies that the design has been correctly synthesized and the FSM transitions are implemented as expected.

13. SIMULATION RESULTS

The simulation confirms the proper sequence of traffic lights.

Time

Road A

Road B

0 ns

Green

Red

20 ns

Yellow

Red

40 ns

Red

Green

60 ns

Red

Yellow

80 ns

Green

Red

The waveform generated in Xilinx shows the LEDs switching correctly according to the FSM states.

14. ADVANTAGES

Simple and reliable digital design.

Low hardware complexity.

Easy to modify timing parameters.

Suitable for FPGA implementation.

High operating speed.

Easy to maintain and upgrade.

Demonstrates practical use of Finite State Machines.

15. APPLICATIONS

Road intersections

Traffic signal automation

Smart city infrastructure

Educational FPGA laboratories

Digital electronics projects

Embedded system learning

Intelligent transportation systems

16. LIMITATIONS

Uses fixed timing and does not adapt to traffic density.

Does not prioritize emergency vehicles.

Does not include pedestrian crossing control.

Requires additional sensors for adaptive traffic management.

17. FUTURE SCOPE

AI-based adaptive traffic control.

Vehicle density detection using image processing.

Emergency vehicle priority system.

IoT-enabled smart traffic monitoring.

Integration with cloud-based traffic management.

Pedestrian signal control.

Solar-powered traffic light systems.

18. CONCLUSION

The Traffic Light Controller Using Verilog HDL successfully demonstrates the implementation of a Finite State Machine (FSM) for automatic traffic signal control. The design was developed using Verilog HDL and verified through Xilinx ISE/Vivado simulation. The controller ensures an orderly sequence of Red, Yellow, and Green lights for two intersecting roads, making it suitable for educational purposes and FPGA-based digital design projects. This project strengthens the understanding of digital logic design, hardware description languages, and FPGA implementation while providing a foundation for future smart traffic management systems.

19. REFERENCES

Xilinx Vivado Design Suite User Guide.

Xilinx ISE Design Suite Documentation.

Samir Palnitkar, Verilog HDL: A Guide to Digital Design and Synthesis.

M. Morris Mano, Digital Design.

IEEE Standard for Verilog Hardware Description Language.

FPGA and Verilog tutorials from Xilinx and academic resources.